PRODUCT USER MANUAL

For
OMI_CLIMATE_TEMPSAL_BALTIC_Ttz_trend

Issue: 4.0

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Approval Date : November 2025

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko	P. Lagemaa
1.1	2019/02/21		Revision	U. Raudsepp I. Maljutenko	P. Lagemaa
2.0	2019/10/10		Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Minor changes	P. Lagemaa	V. Huess
3.0	2023/09/01	all	Template change+ minor changes	P. Lagemaa	V. Huess
4.0	2025/05/22	all	Template change+ minor changes	P. Lagemaa	V. Huess

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_TEMPSAL_BALTIC_Ttz_trend. It contains one dataset:

- omi_climate_tempsal_baltic_Ttz_trend

The OMI provides a timeseries of Baltic Sea area averaged subsurface temperature trends which contains single profile of subsurface temperature trends at different depth levels ranging down to 250 m. The trend is found based on Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_TEMPSAL_BALTIC_Ttz_trend
Geographical coverage	Baltic Sea
Variables	subsurface temperature tendency
Available time series	From 1993 to Year-1
Temporal resolution	Integrated
Horizontal resolution	Integrated
Integration depth (for example)	0 – 250m
Format	NetCDF4

b) Production System Description

The horizontally averaged subsurface temperature trend has been calculated using the results from Copernicus Marine Service BALMFC Reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The linear trend has been calculated from yearly time series of horizontally averaged salinities over the Baltic Sea domain (13 °E - 31 °E; 53 °N - 66 °N; excluding the Skagerrak strait). The upper and lower uncertainty bounds are calculated as the 95% confidence intervals of the yearly trend values.

c) Dataset details

omi_climate_tempsal_baltic_ttz_trend (subsurface temperature trend) contains the following variables:
thetao_trend [365*24*3600*1e-3 s-1] long_name: Horizontally averaged subsurface temperature trend standard_name: tendency_of_sea_water_temperature **thetao_trend_upper95p** [365*24*3600*1e-3 s-1] long_name: Upper 95% confidence bound for the horizontally averaged subsurface temperature trend standard_name: tendency_of_sea_water_temperature **thetao_trend_lower95p** [365*24*3600*1e-3 s-1] long_name: Lowe 95% confidence bound for the horizontally averaged subsurface temperature trend standard_name: tendency_of_sea_water_temperature **time** [days since 1950-01-01 00:00:00] long_name: time standard_name: time

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-TEMPSAL-Ttz-trend)